Théophile GOUSSELOT

Graduate Engineer and PhD Student in
microelectronics, computer science and security

(+ 33)7 81 67 37 68
theophile.gousselot@gmail.com
https://github.com/theophile-gousselot
https://theophile-gousselot.github.io
Full mobility
Full driving licence
Born:1998

I am looking for a challenging position that will stimulate me to develop, attack or protect chips for hours.
I am used to collaborating, presenting and teaching with enthusiasm!

Education

- **PhD Student at Mines de St-Etienne** — since **jan. 2022**
  Protection of RISC-V processors against fault injections
  - **3rd best Hardware Demo (among 24)** at IEEE International Symposium on Hardware Oriented Security and Trust (HOST), 2023, USA, California, San Jose
  - **[Paper]** Code Encryption for Confidentiality and Execution Integrity down to Control Signals, HOST, 2025
  - **[Paper] [Patent]** Lightweight Countermeasures Against Original Linear Code Extraction Attacks on a RISC-V Core, HOST, 2023 (in collaboration with Texplained)
  - Developed a Makefile-based environment for running Verilator simulations with assertions, and for launching Vivado FPGA design flows.
  - Taught Computer Architecture and Digital Design to master's students.

- **Engineering School Mines de St-Etienne** — **3 years**
  Award for academic excellence and involvement in associations.
  Microelectronics: ASIC | FPGA | Manufacturing processes
  - Hardware security: Laser fault injection, Electromagnetic Side-Channel Attacks

  Computer science: Algorithmics | Machine Learning | TCP/IP Architecture
  - Software security: Practical attacks (snifing/man in the middle)

- **Preparatory class : Mathematics & Physics** — **2 years**
  Intensive preparation for the national competitive entrance to leading French engineering schools
  - Solved the traveling salesman problem using an "escargot shell" approach.

Professional experiences

- **Assistant Hardware Security Engineer** — **6 months**
  Lightweight Control Flow integrity against FIA
  - Researched the state-of-the-art solutions (Lock step/Watchdog).
  - Developed a BB signature solution for the SODOR core (Berkley) on an FPGA.

- **National RISC-V Contest** — **6 months**
  THALES
  RISC-V core optimization
  - Reduced program execution time without increasing silicon area.

- **Research Internship in Biomechanics** — **5 months**
  Optimization of the violinist's movement
  - Optimized the violinist's movement with the "Optimal control approach" in order to minimize muscular fatigue.
  - Developed a generic "Optimal control" tool (Python/C++).

- **Assistant Method Engineer** — **3 months**
  Schneider Electric
  - Designed a defect detection with 3D Vision and probing equipment.

Volunteering experiences

- **President of a Student Network Association "MINITEL"** — **1 year**
  Network Wi-Fi/Ethernet | Computer science | Events
  - Managed an assorted team of 20 students
  - Middleman between students and school (paid by the school)

Skills

Microelectronics

Vivado Xilinx	Modelsim Mentor Graphic	Verilator Veripool
Virtuoso Cadence	GTKWave Open Source	Synopsys Synopsys

Tools

Vim	Git	VS Code
Latex	Drawio	Inkscape

Programming

Python	C/Caml	C++
Linux	Bash	Makefile
VHDL	S.Verilog	Chisel

Engineering & Research

Scientific literature review

Scientific writing

Getting Things Done (GTD) method

Languages

French - **native** English - **C1**

Hobbies

Sport
Trek | Mountain bike | Kayak
(for long-term trips)

Environment
Climate Fresk | Facilitator

Culture
Stage acting

Science
C Genial contest (Popularization)